

ActInf ModelStream 014.1: Ran Wei, Value of Information and Reward Spec

ModelStream | 1:15:44

Hello and welcome. It's September 13th, 2024 and we're in Active Inference Model Stream 14.1 with Ranwei talking about value of information and reward specification in Active Inference.

There will be a presentation then we'll have a discussion and looking forward to people's comments and questions. Thank you again for joining and looking forward to the presentation.

Yeah sure so I'll just dive into it. Yeah so thanks again for the invitation and so this paper is about the value of information and then some aspects about reward specification in Active Inference but it also touches on PAMDPs.

And so the motivation is very simple so I was simply trying to understand the expected free energy objective for action planning in Active Inference. And the nice thing about the EFE objective which makes it an appealing objective for a variety of people in a variety of domains is that it has a very intuitive decomposition as the sum of the expected value term and epistemic value term.

And then so the expected value term is this term that encourages the agent to obtain reward or achieve their preferred state of the world because it's simply the cross entropy between the predicted future observations given some actions and the preferred observations $P_{\tilde{d}}$.

The second term epistemic value is what makes active inference or make it what makes EFE very interesting. So the epistemic value term is defined as the expected KL divergence between future future posterior so belief about future states given future observations and the predicted distribution of future states. So what this term does is it quantifies how far is the future posterior from the future prior which essentially quantifies the amount of belief update. So if you optimize this term, then it essentially encourages the agent to take actions that would lead to a higher amount of belief update. So a higher amount of information obtained about the environment.

So experimentally, people have found that optimizing this objective led to some very interesting behavior.

So on the left hand side here, we have an example from one of Alex's papers where he compared the what we call the state coverage of reward maximizing agents versus active inference agents in this simple environment called the mountain car, where the state of the agent is represented by its position and velocity. So we see that active inference agents has a much larger coverage of the state space. So it spreads more territory of the state space versus the reward maximizing agents.

So what this potentially allows the agent to do is to learn a better and more comprehensive model of the environment,

which it also turns out that having such a model enables better action selection and actually achieving higher reward.

So on the right hand side, we have an example from one of my papers. So I did a lot of study on modeling human driving behavior. And this is one of the examples where we have a car called the Eagle vehicle driving down the road.

And then on the side of the road, we have a large truck, which potentially obscures the view of a pedestrian.

But we don't actually know whether the pedestrian exists or not exists. But if the pedestrian does exist, we assume

or rather the Eagle vehicle believes that it's going to cross the road. So we don't want to cause any conflict with the pedestrian who's trying to cross the road. So what this Eagle vehicle does as a consequence of optimizing EFV is that it will nudge slightly to the left of the road to gain view of the pedestrian.

And if it sees that the pedestrian doesn't exist, then it will return to the center of the road and carry on with the optimal

speed without ever having to stop for the pedestrian. So these behaviors are very interesting.

And it highlights the value of encouraging agent to explore and gain information about the environment.

And this is usually understood as the main differentiator between active inference and some of the other frameworks for decision making, such as reinforcement learning. But one somewhat contradictory fact is that actually Bayesian and meta-reinforcement learning are already known to ultimately trade off exploration and exploitation. So here we have an example from one of the more recent meta-RL surveys where, um, uh, so this is actually a very common benchmark in meta-RL where we would start an agent in the center of the circle and in the goal position, which is unknown to the agent will be somewhere along the circle. But again, it's never known to the agent. So the, what the agent has to do is to

explore along the circle to find the goal. So basically what it does is it'll pick a position

along the circle and it will just trace along the circle until it arrives at the goal position. So this is the,

uh, the optimal strategy for this setting. And we see that it's already being demonstrated by meta-RL.

So this begs the question even more of what is the relationship between active inference and Bayesian-RL.

So, um, and the main idea of the paper, or rather the main insight that we're trying to offer is that EFE can be understood as an approximation to the base optimal RL policy and specifically by the information reward shaping of the EFE objective. So to unpack this, I will, uh, do a very brief review of, uh, some preliminary, preliminary annotations on MDPs and active inference. And then we will introduce a few analysis tools from MDP theory, which will allow us to better understand and appreciate the notion of value of information, which then will allow us to derive the main results of the paper. And then we'll just close up with some discussion and more, and in some other interactive activities. Um, yeah, so,

so the preliminaries will be very brief because most people, uh, in the Institute, I believe, are the

experts in these topics. Um, so this is the, uh, regular formalism for MDPs. And then, um,

the objective of an agent in MDP is usually to maximize discounted, uh, cumulative rewards. So I have chosen to use the infinite horizon, uh, setting with discounting because it makes the analysis a bit easier.

But one, so the, so the only thing I wanted to highlight on this page is that, uh, the classic MDP theory says that for a given MDP with known transition probabilities and known reward function, there always exists at least one deterministic stationary Markovian policy. And then the value

function of the policy, which is the value accrued by the policy in MDP, can be computed from the Bellman equations. Um, and then, so another ingredient that we need is going to, is partially observable MDP,

which is just an extension of MDPs, uh, where, uh, the state variable now is no longer directly

observable to the agent, but the agent can observe another signal O that's usually correlated with the

states. Um, but because the agent can no longer observe the state, then if you condition your policy only on the most recent observation, you don't really have, um, sufficient information for, uh, taking optimal actions. So in general for POMDPs, the, uh, optimal policy has to be conditioned on the entire history, which, uh, as accumulates can tend to an infinite length. Um, so here we need to, uh, recall another very, uh, important, uh, result from POMDP theory, which is that the belief state, which can be computed from the Bayesian belief update, is a sufficient statistic of the entire history, which means that if you condition your policy on a belief state, then you don't have to condition your policy on the entire history. And then once you do that, you can define the value function also in terms of the belief. So we have what's called the belief action value function, the belief value function. Um, so this belief state view of POMDPs is very interesting because it says that for any given POMDP, you have an equivalent, uh, MDP that's defined on the space of belief, uh, beliefs, uh, equivalent in the sense that it would yield the same optimal policy for the belief MDP and the POMDP. And then, uh, this belief MDP would

have its corresponding belief action reward and belief transition dynamics. So the belief action reward is defined simply as the weighted average of the state action reward by the beliefs. And what's interesting about this belief MDP is how the belief transition dynamics is defined. Um, so you can understand this by looking at this, uh, these two terms on the right-hand side. So this basically says that in order to transition, um, the belief from one time step to the next time step to get B' , the first thing you're going to do is to draw, uh, a future observation, um, or sometimes I call it counterfactual future observation O' from the posterior predictive of the observation distribution under the current belief. And you're gonna, you, you then you're going to perform a Bayesian update on this counterfactual observation to get, uh, the next belief. And so because the marginal distribution over the observation is in general stochastic, this makes the belief transition dynamics also stochastic. Um, so this is the belief MDP, the equivalent belief MDP for the POMDP. And then we, I will refer to this as the base optimal belief MDP because the policy computed from such a belief MDP would be, would be the base optimal policy for the POMDP. Um, so then we have active inference, which similar to the canonical POMDP framework has an inference step and a planning step. And then, so the difference between active inference and POMDP and the regular POMDP is instead of performing, um, exact Bayesian update, we'll perform variational updates. But, um, because from the variational inference literature, we know that if the variational distribution Q is chosen appropriately, then it would have to be equal to the exact Bayesian update. So this doesn't really introduce too much difference from the classic, um, POMDP framework. The main difference between active inference, uh, and the classic POMDP is that the agent

would, um, instead of optimizing reward, the agent optimizes expected free energy. Uh, and, uh, I, I wrote the, uh, this form of EFE here, which I believe is more general. Um, sometimes it's sometimes referred to as the energy entropy, uh, decomposition. Um, so something I want to remark on this page is that whenever we talk about EFE, we need to be very precise about the definition of different Q s, because there are too many Q s, uh, in active inference and in EFE. Um, and here I define the Q over future states

as the product of, uh, uh, future marginal state distributions. Um, this is not because I invented it. I'm just, uh, writing this, um, uh, writing this, the usual, the, the usual definition of, uh, EFE in a more formal way and explicitly call out that this is the product of future marginal states. Um, and then, so one more step to get the pragmatic and epistemic value decomposition of EFE is that we would have to define the energy function or the preference distribution to have preference only on the observations. And then we can split up the energy entropy decomposition into the pragmatic and epistemic value decomposition. Um, so what this characterization allows us to do is that we can now see EFE as the specific type of belief MDP where, um, it, it would be different from the Bayes optimal belief MDP, but it would still have its own belief action reward and belief transition dynamics. So if we look at the belief action reward, it's simply the sum of the one-step pragmatic value, which I denote with R -tilde data and the one-step epistemic value. And if we unpack the one-step, uh, pragmatic value, we can actually write it as a linear combination of a state action reward. And this is because of the linearity of the expectation, uh, which is used to define, uh, the pragmatic value. Um, but the linearity doesn't hold for the epistemic value. And in fact, the epistemic values concave in the beliefs. Um, another interesting aspect about the belief transition dynamics for EFE is that it doesn't, so different from the Bayes optimal belief transition dynamics, it doesn't contain the counterfactual observation O prime. So what it means is that, uh, instead of drawing a counterfactual observation from the current, uh, posterior predictive marginal and doing a counterfactual update, you would just directly propagate your current belief to the next, uh, state marginal. And so this, uh, is sometimes called the open loop belief updating in a literature in a way that's analogous to open loop control, so that you would update your belief without taking into account of the next observation, but only during planning, not during execution, which we'll come back to later. Um, but the belief MDP view of EFE, uh, would tell us that, uh, there exists an optimal deterministic Markovian policy simply due to the characterization as an MDP. Um, so this, with all of these, um, precise characterizations, we can also make our main question, uh, more precise.

Um, and so we saw that EFE uses the open loop belief dynamics with an added information reward, whereas the Bayes optimal, uh, belief MDP uses the regular, uh, cannot, the regular linear combination of state action rewards and a closed loop belief dynamics. So can we understand, uh, these modifications in EFE

as an approximation to the Bayes optimal belief MDP and in turn the Bayes optimal pulse? So that's the question

we're trying to pursue, uh, in this paper. Um, and then, so in order to answer that, we need to introduce some tools from

MDP theory. Um, so the first set of concepts we'll introduce are the policy advantage and the model advantage.

And for the sake of discussion for now, we'll just understand them as some abstract notion of error measures of, uh, the policy and the model. And the reason why I say that is that if you look at the definition of the policy advantage, for example, so it's defined as the difference between two value functions.

So for the first value function, the action is, uh, taken from the argument of, uh, the advantage function,

whereas in the second value function, the action is taken from, uh, this policy π_i , which is used to define the value function. So the reason why I say this can be understood as an error measure is that if the action a is more or less the same as the action drawn from the policy π_i , then the advantage function will be zero. So there will be no difference between, between them. So this essentially characterizes how much is action a different from an action that would have been chosen by, uh, the policy that's used to define the value function. And then basically more or less the same idea holds for model advantage. But instead of measuring the difference in the actions, we're measuring the difference in the two dynamics by saying that if I draw a next state S' from dynamics P' and another next state S'' from dynamics P , how is the value of S' different from S'' ? So if P' is actually equal to P , then there is no model advantage. So this thing measures the difference between the two models.

Um, two other concepts we need to know, uh, one is effective horizon. So this is basically introduced, uh, uh, as an analog of, uh, planning horizon for infinite horizon discounted problem.

So it's defined as one over one minus the discount, uh, discount factor γ . So it's basically saying how many time steps do I need to accumulate until the geometric discounting that's usually used in, uh, uh, in the RL setting, uh, is zero. And then the normalized occupancy measure is essentially the, uh, stationary for marginal state action distribution along the Markov chain of rolling out a policy in a dynamics.

So this kind of essentially characterizes the equilibrium behavior of the system.

Um, and then so with these concepts, we have our first analysis tool, uh, which I call performance difference in mismatched MDPs. So what we're trying to do in this lemma is the following.

So we're, we say that we're going to look at two policies, π_i and π_i' , and then both policies are optimal with, with, with respect to their own MDPs, which we denote as M and M' .

And we, we're going to say that these two MDPs would share the same state and action space, same initial state distribution, and the same discount factor, but they could potentially have different rewards and different dynamics.

So this is, this, this is very similar to the comparison of, uh, the EFE policy and the base optimal policy where, um, they share the same spaces,

but they have different rewards and different dynamics. So this is basically our tool to quantify their difference.

And then, so what this lemma said, move on to say is that, um, we're going to look at the performance difference between these two policies,

but both of them will be evaluated in environment M .

And because, uh, policy π_i is by definition, the optimal policy for, uh, MDP or environment M , then there's no way that π_i' is going to do better than π_i .

Uh, so that's why this quantity is also called the regret.

Uh, so we're, we're going to interchange with calling it regret or performance difference.

But, um, basically this relationship here shows that the regret can be decomposed into three terms.

So the first term is the advantage of the actions drawn from, uh, uh, policy π_i ,

which I would, uh, I call it the expert distribution because it's the better policy in this setting.

And then the second and third term, um, are defined in terms of the model advantage.

And I've used, um, notation or abuse labeling to call it also the reward advantage.

And then, so you can understand this relationship, uh, by, um, saying that if R and R' are equal, and then P and P' are equal, then there's no reward model advantage or disadvantage between the two policies.

And because of that, then there's also no policy advantage, then the regret would be zero.

Um, so, and, and so the nice thing about this relationship is that if we put an absolute value sign, uh, over it, and then we, uh, do some algebra, then we can actually obtain an upper bound on the regret.

That's defined in terms of the policy advantage, reward advantage, and, uh, the difference between the two dynamics.

So from now on, I will actually call reward and policy advantage, um, reward and policy error, because that's to some extent what they represent.

So what we see here on the right hand side is that the coefficients for these different terms are different.

So the coefficients for the reward and policy error are linear in the planning horizon, one over one minus gamma,

whereas the coefficient for the, uh, dynamics difference is quadratic or squared in the planning horizon.

So it says that if we're, uh, making errors or mistakes along all of these dimensions, then the rate at which the dynamics model error accumulate is going to be much faster than the rate at which, uh, policy and reward error accumulates.

Another thing to remark about this relationship is that it can be a little bit conservative because, um, sometimes even if you make mistakes with your choice of dynamics,

you can actually make it up by, uh, choosing the reward function to be different from the optimal reward.

So basically whatever, if, if you incur some disadvantage in the dynamics, then you can choose to gain some advantage in the reward so that it can cancel out.

And so basically I'm going to show that this is what happens with, uh, the EFE policy, EFE policy compared to the base optimal policy.

Um, yeah, so that's with this, let's refine our main question a little bit more.

So we're trying to understand EFE as an approximation to the base optimal, uh, belief MDP.

And then specifically the way in which, um, it approximates that is via this reward shaping term to cancel out the disadvantage, uh, due to using open loop belief dynamics.

So before we do that, we, um, we should review the notion of value of information, uh, which will give us a clue, uh, and also make better connections between, uh, these two topics.

So the value of information, uh, the most popular definition was given by Ronald Howard 1966, where he defines it as the reward that a decision maker is willing to give away.

So, um, if they could have their uncertainty resolved, um, by some magician, for example, and if you commit to this definition, then you can quantify the value of information simply as the difference between the expected value given perfect information.

And then one very appealing fact about this is that because an optimal decision maker cannot do worse by

it's due to the disadvantage of using open loop policy,
open loop dynamics and open loop policy planning.
So what exactly is the advantage
or disadvantage of being open loop?
So here we have a proposition that quantifies that.
And it turns out that the advantage of being closed loop
is exactly related to the information gain.
So how did we get this?
So basically we start with a one step decision making.
And in that setting,
the advantage of being closed loop
is basically the expected value of perfect observation,
which is the difference between the expected value
given perfect observation and without.
And then we can rewrite this as a linear combination
of the difference between the updated belief
and the prior belief.
And if we apply some balance here,
then we can rewrite, we can upper bound this
as the expected information gain.
And that's how we got this.
And if we accrue this along the planning horizon,
then we basically multiply this
by the effective planning horizon,
one over one minus gamma.
So that's how this is obtained.
So this is actually very interesting
because it gives us a clue that
if we want to improve the open loop policy,
while still committing to open loop dynamics,
then one very reasonable thing we can do
is to change its reward.
And a very obvious choice for changing its reward
is to simply add,
is to simply make up for the information gain
that it loses by using the open loop dynamics
and add that back to the reward function.
And once you do that,

then basically you got the EFE objective function in here.

So to be able to formally show that the EFE policy,

which is simply the open loop policy

plus an information reward,

and show that how much it improves

on top of the open loop policy,

we need to make a few assumptions.

So the first assumption we make here

is on the preference specification,

so basically we need to,

what this assumption says is that

we need to make sure that

if we still want the EFE agent to be goal-directed,

then we need to set the preference in such a way

so that the gain in pragmatic value

after a belief update

is always greater than the loss in information gain

after a belief update.

So this way we make sure that the agent focus on a task

and doesn't get distracted by collecting information

that's potentially not useful for the task.

And then the second set of assumptions are introduced

to make us able to compare the open loop policy

and EFE policy on more of equal ground.

So the first assumption is that

the, we're going to assume that the EFE policy commits

no more, or rather it does no worse

in terms of policy advantage

compared to the open loop policy.

And the second assumption says that

the information gain value along the horizon

is always going to be greater than some values.

So again, these things are introduced just for us

to better compare the two policies.

And with all of these assumptions

and the previous insights, we have our main result here,

which gives the regret bounds for the open loop

and the EFE policies.

And so we see here that for both the open loop and EFE policies, we have a term that's linear in the policy error and quadratic in the information gain.

But the difference between the EFE policy and the open loop policy is that the EFE policy has an additional term, which is a minus something that is linear in the information gain.

So basically the upper bound is being reduced by this linear, this term that's linear in the information gain.

So that's how much the EFE policy improves upon the open loop policy, which makes it a better approximation to the base optimal policy than open loop policy.

Yeah, so that's the main results and the main idea in the paper.

So we'll just close up with some discussions that might be interesting to people from different communities.

So the first interesting observation is that this view of EFE as an approximation to the base optimal policy actually connects back to a lot of work people have done in POMDP planning.

So the thing about POMDP planning that's very tricky is that even though MDP policies are very easy to compute, as soon as we introduce partial observation, it just becomes extremely difficult to compute the optimal policy that would optimally trade off exploration and exploitation.

So people have explored a bunch of heuristics and one of them is called the dual mode controller, where you would design an ad hoc rule to switch between exploration mode and exploitation mode.

So for the explore and that switch is being determined by the entropy of your belief with some chosen threshold.

And so for the exploration mode, you essentially take the argmax of the current belief.

So that's the state that you believe is most likely to be the current state.

And you would simply choose an action from the MDP policy,

which we know is easier to compute.

But if the belief entropy is higher than your threshold,
then you would take an action to minimize the entropy of the next belief.

So you can read this as something that's very similar
to maximizing the gain of information in the next period.

Another interesting framework, again, from the POMDP literature,
is that some people have realized that there are just,
there's certain types of behavior that just cannot be modeled
as obtaining, as reward driven.

And usually what happens here is that you want these agents
to actually minimize the entropy of their beliefs
or maximize the information gain.

So this is usually called RO-POMDP or active sensing.

So the objective function in this framework
usually is just to minimize entropy of belief,
but not the one step belief entropy,
rather the cumulative belief entropy
along the entire planning horizon.

So something very interesting about this is that
this term, if we introduce
a uniform prior belief that doesn't change over time steps,
then minimizing belief entropy is actually proportional
to maximizing the expected KL divergence
between the updated belief and a uniform prior belief.

So this looks really similar
to the information gain objective in EFE.

And usually what people do is that instead of just minimizing belief entropy,
depending on whether we have any requirements or getting rewards,
we would do a weighted sum of the reward term
and the belief entropy term.

And when you do that, that objective is essentially the belief action reward
in the EFE case.

So the one final thing is that actually goes back to active inference.

So the formulation of active inference that I have introduced,
I call it vanilla active inference.

And it's also the active inference,
the version of active inference that's currently in the main branch of MDP.

But in 2021, people have proposed a variation of active inference called the sophisticated inference.

And basically what it does is that it maintains the reward plus information gain as the belief action reward.

But it actually uses the closed loop belief dynamics.

I think there's a sentence somewhere in the paper that says that the difference between sophisticated inference

and the vanilla active inference is that the agent would plan not for future states, but for future beliefs.

So if that's the case, then it's essentially using the same belief transition dynamics as the base optimal belief MDP.

So for practitioners, then there's a potential problem with sophisticated inference, which is that now you're basically being based optimal with your belief dynamics,

but you're adding an additional term in your reward, which means that if the behavior that's supposed to be produced by the base optimal policy,

you are now ruining it with the additional information gains.

So basically the concern is that agents, sophisticated inference agent would try to get too much information at the detriment of getting less reward,

because you would spend more, basically you would spend more time collecting information than focusing on a task.

But essentially what comes to the rescue is our old friend, the temperature parameter.

So depending on how you set the precision of your preference distribution, for example, using a temperature parameter of a softmax distribution,

then you essentially construct weighted, essentially a weighting of the reward term versus the information gain term.

So if your preference is very precise with a very low temperature, then in this case, a very high value of λ ,

then you're essentially saying that I only care about the reward. I don't care about the information gain.

So if you do this, then sophisticated inference just becomes exactly equivalent to the base optimal policy.

So essentially the λ , temperature term λ , interpolates base optimal policy for reward seeking behavior versus information seeking behavior,

if you have low λ . And this essentially makes sophisticated EFE, a strict generalization of the base optimal policy.

So that's everything I wanted to discuss. I think, so the final slide is for people who are interested in more the basics of active inference

and also the connection between active inference and RL. I have two blog posts.

I have two blog posts to offer that some people perhaps find helpful.

Yeah. Thanks again.

Awesome. Thank you.

All right. I'll read a first question from the live chat and then I'll read some questions I wrote and some other questions from the live chat.

So Andrew wrote,

You mentioned that we should set preferences such that the pragmatic value outweighs the information value lost after the agent has already achieved that information value so that the agent does not become too

focused on seeking information rather than achieving a goal.

Can this strategy risk the agent getting stuck in a local optimum by setting preferences which mitigate information seeking?

Okay, I think something slipped my mind because it appears to me that the question is somewhat reversed.

So if I'm getting the idea right, so you're saying that if the agent somehow gets stuck in a local minima along a trajectory to get reward,

then one thing you can do is to change, for example, the weighting between the reward term and the information gain term.

So that it essentially switches the mode to focus on getting information and hopefully balances itself out of the local optimum.

Is that right?

It could be.

And then is the question whether that's a reasonable thing to do, whether there's a way to automatically do it?

Or is that what you're looking for?

I don't know.

Yeah.

Where and how does preference learning and scaling come into play to keep these kinds of epistemic and pragmatic dualities?

If there is going to be a unified imperative for policy selection that's projecting onto a set of EFE values, then how do you weight the epistemic and pragmatic value in training and in deployment?

Yeah, I think this is an open question.

I don't think anyone has done very convincing work on this topic specifically.

So I know about one thread of work from Carl's people.

Essentially, they have a way to define an update rule for the temperature parameter or the precision parameter.

So basically what happens there is that if you end up in a situation where your variational free energy, which quantifies the accuracy and essentially the accuracy of your predictive model, if variational free energy decreases,

which means that you're starting to observe signals that constantly violates your model of the environment, then you would choose actions in a way that's more stochastic.

So there are ways to derive such rules from the view where of this notion of free energy unifies perception and action.

So people from Carl's lab has derived rules like this.

And then there's also work from Noor, and I think it's called,

I can't remember what it's called,

but basically what that work does is it's trying to use Bayesian updating to learn the preference as the agent interacts with the environment.

So basically the agent would start by having some preferences

about what states or observations to achieve in the environment.

And as the agent interacts with the environment,

it would actually refine its own model,

its own preference model,

rather than its model of the dynamics of the environment.

But something very tricky that happens there is that,

so she did the experiment in this environment called frozen lake.

So basically you're walking on a frozen lake and there are holes on a lake.

And if you somehow step into that hole,

you would fall or die.

And then an agent that learns the preference in that environment

basically learns that it prefers to fall into holes.

So that doesn't really make a lot of sense.

If your goal is to maximize reward or stay in the life,

but it makes sense in a sense that agent just builds a preference model

that is essentially in compliance with what actually happens in the environment.

So it's kind of self-evidencing in the wrong way.

So do either of these two approaches provide a practical way

to automatically tune a temperature parameter

to balance yourself out of a local optima, like you said?

I don't think either of them do that.

So I don't think I have a comment that firmly addresses this at the moment.

I do know that people who actually work more on the Bayesian RL side,

they have done some work to derive approximations to exact Bayesian RL.

And in doing so,

they were able to derive some automatic temperature tuning rules to achieve that.

So if you're interested, I believe the person's is Ian Alsbland.

But I can be wrong.

But just look for Bayesian RL literature.

It's a paper that essentially derives a Bayesian RL version of soft active critic.

Awesome.

Thank you.

And another way to address that is to avert slash avoid structurally

so that like a blood sugar distribution could be loaded with high pragmatic value.

An I-secade distribution could be driven by high epistemic value in hierarchical models.

So it isn't that every single model is just loosely describing all of this cluster of undefined processes.

This is really getting to the distributional level.

And then if a distribution really does have such a challenging situation

that it has to be tight and refined and then make a large jump to an area it's never been.

I mean, those are the challenging situations.

And that's where these other techniques and methods can be brought into play.

However, those are the challenging ones.

And that is not even necessarily needed for simpler settings.

Well, thanks again for the really instructive and informative presentation.

To kind of just restate, I saw the big move as grounding in the full observability conditions and the analytic situation for the Markov decision process for the MDP.

So certain kinds of guarantees exist that are, as they should be, slightly weakened with the partial observability introduction of observables and latent states.

So how is it possible to use some of the techniques, patterns, guarantees, etc. for full observability on a partially observable situation?

And the move is kind of to consider the parameterization of the generative model as a fully observable setting, which is what it is in the programming environment.

Whereas if even part of it were obscured, then even that would be a partially observable setting.

So by treating the ABCDE partially observable MDP that we see in the textbook and all this work, by treating that as fully observable, essentially to the proof finder, exactly, thank you,

by treating the full state updating of the POMDP, now the bottom-up semantics of those variables might be the probability that this happens or the probability that this maps to that.

So still it is about partial observability, but then considering that object in its fully observable condition to itself, then you can see that no-regret situation where there's no policy advantage and one's own sensemaking didn't have a disadvantage and an expert evaluating one's sensemaking didn't have a disadvantage.

So in those situations, the learner's heuristic is zero regret on the policy and on the sensemaking side. And then there's a kind of imperative, which is the distance from the expert sensemaking

and policy selection.

Because it's basically like,
what would have been the expert policy and sensemaking
if it would have been fully observable?

And then the POMDP reduces to an MDP.

Yeah, that's precisely the strategy.

So planning in POMDPs
can be turned into planning in MDP.

It's just a much more difficult MDP.

You don't get away with any of the difficulty
of planning in a belief MDP

because now the state space
essentially has countably infinite cardinality,
which means that there's no way

for you to compactly represent
your beliefs over multiple steps
or your value functions.

So it doesn't make any of that
or any practical applications simpler,
but it does offer a very nice theoretic technique
to better understand the different kinds of tweaks
you can do to the theoretically optimal,
base optimal policy.

Cool.

And then another connection
is with the dual mode switching.

So in like chapter five of the textbook,
in their neuroanatomy,

they show one schema
of how dopamine level

regulates sort of a system one,
system two,

thinking fast, thinking slow shift.

In the thinking fast mode,
observations expected or otherwise
don't even need to be considered.

You can just sample directly
from the policy prior,

from habit.

And then even to engage
expected free energy,
use G to update the policy prior
into a policy posterior.

So even to consider
possible future actions
and observations,
that's a deliberative mode.

It's dealing with intangibles
and doing computations
that where habit will suffice
can just be done.

So then by proxy,
you can use the observation entropy
and just say,
if the observations are within
this acceptable threshold,
there's no reason to step outside
of habit.

And then where observations
start to become surprising
in this way,
then a deliberative execution
per policy can happen.

Yeah, so I think this is very interesting.

So mode switching
or essentially self-selection
of how much compute resources
you put into at test time
or at runtime
is a very interesting topic.

I know Alec has done some work
on like once you amortize a policy,
you can still choose
to refine your policies online
and so on and so forth.

And it also makes a nice connection

between what's happening
with like large language models.
Now that you have like so-called
like post-training,
you can do a lot of stuff.
So for example,
in what's called best of end sampling,
where you sample a ton of outputs
from the policy
and then you use a reward function
to score which one is better
and then choose the output that
instead of all the other completions
that you have sampled.
And then basically,
the more completions you sample,
the more likely you'll get a completion
that's going to be scored higher
by the test time reward function.
And then you can basically decide,
okay, what's the trade-off
I'm going to make
between my test time compute
that's been used to do a ton of sampling,
which is very expensive,
versus the potential improvement
in reward that I'm going to get.
So this is very interesting.
But I don't think
like in the active inference community,
people have done enough principled work
or come up with a principled framework
to allow an agent
to dynamically adjust
their computational resource
and time on the fly.
So I would actually encourage,
and also I don't really understand it myself,

so I would encourage people
to do more work
along this direction
to figure out
what is the generative model
that allow people
to do test time adaptation
of compute resources
and so on.

Cool.

Yeah, that's very interesting
and gets into like system design.

Like, is it the agent
that's estimating
how much resources
would be useful for it?
Or is there an estimator
that's making a secondary judgment?

I'll read a question from the chat.

Sriharsha wrote,
thanks for the talk, Rand.

In your expertise
from an engineering perspective,
what are your comments
on building a deep
active inference agent
versus a deep RL agent
like Dreamer?

Yeah, so I think
the word deep
sometimes mean different things.
in different communities.

So in,
so early in the days,
deep active inference
used to mean
using like deep networks
to approximate

certain distributions
or amortize certain distributions
in active inference agents.

I think over,
and then over the years,
the meaning of deep
has shifted.

So now I think
within the active inference community,
deep means
means having
multiple levels
of hierarchy.

Sometimes it also means
planning a long,
a longer
temporal depth.

Yeah, so,
so I think,
I think
being hierarchical
is actually pretty difficult.

I, again,
I don't know of a very
nicely principled
and still
working way
to make agents
hierarchically deep
and also
it's first principle
in the sense
that it can be
automatically discovered
from data.

I've tried to do some work,
but I haven't
found a way

in which
the hierarchy
doesn't
collapse.
Usually what I've found
is hierarchies
tend to collapse
and things actually,
like anything
that you can
portray as hierarchical
can also
be
represented
in a flat way.
So I would,
again,
encourage people
to do more
work in that direction.
but in terms
of just comparing
the various
tools
and also
way to hybrid
active inference
with,
with neural nets
that people have done
versus agents.
So I think
basically
a lot of people's
view of Dreamer
versus,
you know,
deep active inference

that many people
have implemented
is that
Dreamer
is
so similar
to active inference,
especially,
I can't remember
which version
of Dreamer
and if it does
use an information
gain term
in a planning objective.
And if it does,
then
you basically
cannot tell
whether there's
any difference
between Dreamer
and deep active inference.
So,
so in a sense,
even
Carl has mentioned
sometimes that
Dreamer
is basically
like
some
variations
of an active
inference agent.
I would,
I would point,
but I would still

point out
that there's
one aspect
of Dreamer
which makes
it a little bit
theoretically
suboptimal,
but it might
work well
in practice
because,
because of
that suboptimality
is that
Dreamer
doesn't perform
belief-states planning
as most
deep RL systems
don't perform
belief-based planning.
They perform
state-based planning
because if you look
at the
policy architecture
of Dreamer,
the policy
is not
conditioned
on a
representation
of belief,
it's conditioned
on a single
or multiple
particles

sample from
the posterior.

So that
in theory
would lead
to a
slightly
different
exploration
behavior,
but I'm
not sure
how that
actually behaves
and the
difference
it makes
in practice.

So if you're
interested in that
and if you want
to make a deep
RL agent
actually do
belief-based planning
which I personally
have some
interest in,
I would encourage
you to look at
some of the work
from
Shimon Whiteson's
lab
from Oxford.
They have done
a lot of
belief-based

planning
of deep
RL agents.
So for people
who,
I think
some people
in the institute
might know
this,
there's an
algorithm
called
VeryBAT
that
the active
influence folk
has cited
multiple times
so that
essentially
came out
of
Shimon Whiteson's
lab.
And in
that architecture
the agent's
policy is
conditioned
on a
belief
representation
as a
supposed
to
state
representation.

Awesome.

Thank you.

Yeah.

A kind of
short comment

and thought

on this

theory and

practice,

like you

mentioned

multiple

namespaces

that collide

with deep

active

inference.

Sometimes

people are

using neural

network

approximators,

sometimes

that's called

amortized

inference.

And then

there's

sort of

in terms

of model

depth,

people could

be talking

about

hierarchical

like

hierarchical

Bayesian
models.

They could
also be
talking
about
temporal
depth.

So whether
it's
more
layers
of the
cake
on this
time
slice

or whether
it's
however
many
layers
of the
cake
with
more
time
slices.

And then
you kind
of brought
in a
twist
with
however
hierarchies
can be
flattened.

We see
that all
the time
in matrix
operations
where
sub
matrices
are used
to
flatten
all kinds
of things.
And then
similarly
in
planning
there can
be a
thicker
presence
or
habit
and
heuristics
can be
learned
such that
planning
like
behavior
can be
emulated
with
less
depth.
So
that's

just so
interesting
like
the
fact
that
these
formalisms
can
go
and
deep
in
space
and
time
is
really
just
like
an
initial
surveying
of
spaces
and
just
saying
within
the
infinite
policy
horizon
situation
this
and this
and this
will be

the
case
and
then
it's
a
purely
empirical
engineering
question
whether
time
horizon
of
1
2
3
4
5
has
what
computational
trade-offs
and needs
so
it's
just
like
so
amazing
how
the
math
has
already
sketched
out
some

of the
dimensions
and
then
within
the
space
of
what
is
sketched
out
it's
an
engineering
question
almost
alone
and
there's
still
theoretical
differences
to clarify
this
entangle
but
even
those
alone
don't
map
obviously
to
performance
adequacy
or
advantage

in the
real world
or for
any
given
data
set
so
it's
kind
of
like
fun
to
research
and
discuss
this
underlying
layer
which
gives
rise
to
those
engineering
results
at
the
same
time
though
imagining
that a
certain
graph
or
equation

has
a
certain
output
on
a
certain
data
set
is
hopeful
and
then
also
I
think
it
connects
to
the
low
and
the
high
road
like
with
different
architectures
that
recapitulate
and get
to
their
own
syntheses
and
combinations

of
what we
call
pragmatic
and
epistemic
value
or
maybe
what
in
the
energy
based
learning
energy
based
models
that
they
might
use
more
like
you
presented
with
energy
and
entropy
decomposition
it's
like
those
are
constructions
along
the

low
road
they
are
model
architectures
that
get
to
this
common
nexus
of
being
able
to
have
an
imperative
which
unifies
both
epistemic
and
pragmatic
value
the
free
energy
principle
coming
from
the
high
road
on
the
other

hand
has
no
architectural
suggestion
it's
a
partition
in
the
mathematical
physics
state
space
and so
it's
unsurprising
that there
are
models
that
are
similar
on
through
further
to
simplified
low
road
constructions
like
POMDP
because
we're
all
talking
about

the
same
setting
agent
or
interface
input
and
output
with
pragmatic
and
epistemic
component
and so
just
interesting
how
all
of
those
architectures
even
though
they
might
not
be
able
to
make
API
calls
to
each
other
or
hot

swap
different
components
they
have
comparability
against
the
high
road
understanding
of
their
belief
updating
for
the
next
observation
coming
in
and
action
going
out
in
a
fitness
context
yeah
I
don't
have
much
comments
on
high
road

stuff
I
personally
would
label
myself
as
a
lower
road
person
that's
that's
awesome
but
both roads
must be
built
yeah
um
I'll ask
another
question
in the
live
chat
and then
we'll
see
if
there's
any
final
questions
upcycle
club
wrote
are

there
any
potential
benefits
of
implementing
sparse
coding
techniques
in
this
framework
hmm
yeah
so
I'm
not
super
familiar
with
that
let me
let me
google
that
real
quick
because
I
do
want
to
see
if
I
can
address
that

question
and
I
know
about
sparse
coding
and I
know
the
connection
between
that
and
VAE
stuff
let me
look that
up
real
quick
okay
so
I'm
looking
at a
tutorial
on
sparse
coding
says
sparse
coding
is
a
class
of
unsupervised

models
for
learning
over
complete
representations
yeah
so
I'm
not sure
if there's
a way
to interact
with
the
viewer
but
does
that
mean
that
you're
trying
to
so
basically
learn
more
representations
than
what's
sufficient
in a way
that's
kind of
similar
to how
people

are
training
sparse
auto
encoders
for
interpreting
language
models
nowadays
I'm
not
sure
that's
maybe I
should
read on
a little
bit
more
to see
what
benefits
it
offered
yeah
I'll
let
you
know
if
any
other
information
comes
in
it's
an

interesting
connection
between
the
matrix
flattening
and the
sparse
coding
because
sometimes
when the
matrices
get flattened
and fused
together
using the
submatrix
representations
then they
have
sparsity
and then
that
actually
just from
an
engineering
perspective
enables
sparse
matrix
methods
and
compression
to be
used
so

again
whether
the
hierarchical
representation
and with
a lower
density
or a
sparser
flattened
representation
for a
given
like
language
etc
has
this or
that
performance
that's
just
an
empirical
question
but
it's
interesting
that
flattening
gives
you
a
constructed
sparsity
even
for

non
sparse
information
and then
there's
the
angle
you
mentioned
about
the
interpretability
sparsity
and
methods
to pull
out
fewer
interpretable
characteristics
like
for example
yeah
did
the
viewer
respond
there
no
but
how about
just
upcycle
better
handling
of missing
data
would be

one
benefit
I can
think
of
so
maybe
we'll
research
and learn
14.2
what
are
your
current
research
or
development
directions
so
I'm
interested
in
pursuing
this
view
of
principal
planning
and
active
infants
a little
more
so
basically
the
reason

why
I
did
the
study
and
wrote
this
is
that
I
was
trying
to
understand
if
there's
way
to
develop
better
planning
algorithms
for
agents
and
specifically
belief
space
planning
for
agents
in
a
more
principled
way
and

initially
I was
doing
some
experiments
and
then
it
occurred
to me
that
I had
doing
experiments
for
belief
space
planning
of
active
inference
agents
and
I
realized
that
if
I
want
to
design
better
algorithms
for
active
inference
agents
I

actually
need
to
have
a
very
precise
understanding
of
what
the
active
inference
agents
is
supposed
to
do
in
a
way
that's
not
susceptible
to
all the
holes
in
active
inference
theory
currently
so
this
paper
essentially
contributes
to

filling
up
some
of
holes
in
the
theory
of
active
inference
agent
planning
but
just
more
like
planning
and
approximate
belief
space
planning
in
general
so
what
I
want
to
do
as
the
next
step
for
the
stuff

presented
in
this
paper
is
to
see
if
there
are
better
ways
to
improve
planning
capabilities
of
active
inference
agents
and
belief
space
planning
for
agents
in
general
that
might
depends
on
better
leveraging
tools
from
variational
inference

for
planning
and
my
leverage
tools
from
temporal
abstraction
and my
leverage
tools
from
essentially
sparsity
or sparsification
or inducing
more structures
in a
generative
model
but
what I
really
want
to
do
is
just
make
better
planning
algorithms
for
agents
any
algorithms
that

actually
work
that
doesn't
require
a lot
of
tuning
that
you
can
just
grab
and
run
it
or
any
ideas
you
have
in
mind
you
don't
have
to
look
for
very
specific
tools
on
the
internet
and
try
it

and
realize
it
doesn't
it
doesn't
really
work
that
well
awesome
I'll
ask
one
last
question
in
theory
or
in
practice
how
is
planning
for
overt
action
embodied
movement
similar
and
different
than
planning
for
covert
mental
action

selection at the higher level of the hierarchy or rather well so basically casting that as covert action selection. So this doesn't really, at least if you approach it from the view of action as interface between the agent and the environment, then in that sense action always have to exist at the lowest level of the hierarchy right at interface between the agent and environment. So it will never be an overt action that exists inside of the agent at the higher level. So that's part of the reason why I say that people should do more work on hierarchical models and hierarchical planning because even though a lot of the existing techniques work, so for example in hierarchical planning we have the options framework from hierarchical RL. So what that does is basically decompose planning at the planning of primitive actions at the lowest level to planning at the higher level over more extended periods of time. So basically the higher level will say execute this task which then gets translated to a sequence of lower level actions such as going to the kitchen, grab coffee, right? That's a single movement at the higher level even though it's a bunch of lower level primitive movements. So I think these techniques people have been using in hierarchical planning are very appealing and they actually work pretty well. And now with like using language models to do the option selection, they work very well in practice. But in theory, I feel like there's still a gap of how hierarchical perception model arise automatically from the environment and how planning can just automatically be a part of that. And then how overt mental actions play a role after you have obtained or learned hierarchical representation of the environment. So all of these, in my opinion, are open questions. I think people need to continue to address them. I don't think people have addressed them to a satisfactory extent at the moment. Cool. All right. Thank you. This was a great stream. So till next time. Yeah. Awesome. Bye. Yeah. Bye.

Thank you.